20/10/26 Week - 3

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~~Ant colony optimization~~

10 Ant colony optimization:

Application: Campus Shuttle Bus Route Optimization

Pseudocode:

Begin

1. Initialize:
   Define campus locations
   Define distance matrix
   define travel time matrix
   define passenger demand
   define bus capacity
   Initialize pheromone values
2. Repeat for a fixed number of iterations:
   For each ant:
      Start from the bus depot
      Mark all locations as unvisited
      Set current load = 0
      Set total distance = 0
      Set total time = 0
      While unvisited locations remain:
         Find feasible next locations
         Calculate selection probability
         using pheromone and heuristic values
         Select the next location
         Update:
            Current location
            Passenger load
            Total distance
            Total travel time
         Mark location as visited

If bus capacity is reached;
Return to depot
Reset passenger load
Return to depot
Calculate total route cost
END FOR
Select the best route
Evaporate existing pheromones
Deposit new pheromones on good routes
3 Display the best route
4 Display:
Total distance
Total travel time
Numbers of buses required
Passenger demand served
END

Formulas:
Cost = $\alpha * D + \beta * T$
D - total distance travelled
T - total travel time
α - Weight assigned to distance
β - Weight assigned to time

O/P:
Enter numbers of locations : 5
#Take all locations input and demand and bus cap
and distance and time b/w two locations
--- OPTIMIZED SHUTTLE ROUTE ---
Trip 1
Depot → Hostel → Library → Depot

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Trip 2
Depot → CSE Block → Parking → Depot

Total Distance : 20.0 km
Total travel time : 23.0 minutes
Total Passenger Demand : 18
Bus Capacity : 10
Number of Trips : 2
Minimum Objective Cost : 43.0

All passenger demands are served.

6/11/26

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```
import random

# ---------------- USER INPUT ----------------

print("CAMPUS SHUTTLE BUS ROUTE OPTIMIZATION")
print("-------------------------------------")

n = int(input("Enter number of locations (including depot): "))

locations = []

for i in range(n):
    name = input(f"Enter location {i + 1}: ")
    locations.append(name)

print("\nFirst location is considered as DEPOT.")

# Passenger demand
demand = [0]
```

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```
# ---------------- COST FUNCTION ----------------

def calculate_cost(route):

    total_distance = 0
    total_time = 0

    for i in range(len(route) - 1):
        a = route[i]
        b = route[i + 1]

        total_distance += distance[a][b]
        total_time += time[a][b]

    return total_distance, total_time


def route_cost(route):

    d, t = calculate_cost(route)

    return d + t
```

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```
print("\nEnter travel time between locations (in minutes):")

for i in range(n):
    for j in range(i + 1, n):
        t = float(input(
            f"Time from {locations[i]} to {locations[j]}: "
        ))
        time[i][j] = t
        time[j][i] = t

# ---------------- ACO PARAMETERS ----------------

ants = 20
iterations = 100
alpha = 1
beta = 2
rho = 0.5
Q = 100

# Initialize pheromone matrix
pheromone = [[1.0] * n for _ in range(n)]
```

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```
for i in range(1, n):
    d = int(input(f"Enter passenger demand at {locations[i]}: "))
    demand.append(d)

capacity = int(input("\nEnter bus capacity: "))

# Distance matrix
distance = [[0] * n for _ in range(n)]

print("\nEnter distance between locations (in km):")

for i in range(n):
    for j in range(i + 1, n):
        d = float(input(
            f"Distance from {locations[i]} to {locations[j]}: "
        ))
        distance[i][j] = d
        distance[j][i] = d

# Travel time matrix
time = [[0] * n for _ in range(n)]
```

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```
# ---------------- CONSTRUCT ROUTE ----------------

def construct_route():

    unvisited = set(range(1, n))

    route = [0]
    current = 0
    load = 0

    while unvisited:

        feasible = [
            j for j in unvisited
            if demand[j] <= capacity - load
        ]

        # Return to depot if capacity is insufficient
        if not feasible:

            route.append(0)
```

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```
        for i in range(n):
            for j in range(n):

                pheromone[i][j] *= (1 - rho)

        # Pheromone update
        for route, cost in ant_routes:

            deposit = Q / cost

            for k in range(len(route) - 1):

                a = route[k]
                b = route[k + 1]

                pheromone[a][b] += deposit
                pheromone[b][a] += deposit


    # ---------------- DISPLAY RESULT ----------------

    total distance. total time = calculate cost(best route)
```

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```
            ]

            next_location = random.choices(
                feasible,
                weights=probabilities
            )[0]

            route.append(next_location)

            load += demand[next_location]

            unvisited.remove(next_location)

            current = next_location

        route.append(0)

        return route


    # ---------------- ACO ALGORITHM ----------------
```

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```
print("\n========== OPTIMIZED SHUTTLE ROUTE ==========")

trip = 1
passengers = 0

print("\nTrip", trip)
print(locations[0], end="")

for i in range(1, len(best_route)):

    node = best_route[i]

    if node == 0:

        print(" ->", locations[0])

        if i < len(best_route) - 1:

            trip += 1
            passengers = 0

            print("\nTrip", trip)
```

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```
best_route = None
best_cost = float("inf")

for iteration in range(iterations):

    ant_routes = []

    for ant in range(ants):

        route = construct_route()

        cost = route_cost(route)

        ant_routes.append((route, cost))

        if cost < best_cost:

            best_cost = cost
            best_route = route

    # Pheromone evaporation
```

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[1] 2m

```
            route.append(0)
            current = 0
            load = 0
            continue

        probabilities = []

        for j in feasible:

            eta = 1 / (distance[current][j] +
                       time[current][j] + 0.001)

            value = (
                pheromone[current][j] ** alpha
            ) * (eta ** beta)

            probabilities.append(value)

        total = sum(probabilities)

        probabilities = [
            p / total for p in probabilities
```

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```
            print("\nTrip", trip)
            print(locations[0], end="")

        else:

            print(" ->", locations[node], end="")
            passengers += demand[node]

    print("\n\nTotal Distance:", total_distance, "km")
    print("Total Travel Time:", total_time, "minutes")
    print("Total Passenger Demand:", sum(demand))
    print("Bus Capacity:", capacity)
    print("Number of Trips:", best_route.count(0) - 1)
    print("Minimum Objective Cost:", best_cost)

    print("\nAll passenger demands are served.")
```

```
CAMPUS SHUTTLE BUS ROUTE OPTIMIZATION
--------------------------------------
Enter number of locations (including depot): 5
Enter location 1: Depot
Enter location 2: Hostel
```

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```
Enter location 3: Library
Enter location 4: CSE Block
Enter location 5: Parking

First location is considered as DEPOT.
Enter passenger demand at Hostel: 6
Enter passenger demand at Library: 4
Enter passenger demand at CSE Block: 5
Enter passenger demand at Parking: 3

Enter bus capacity: 10

Enter distance between locations (in km):
Distance from Depot to Hostel: 2
Distance from Depot to Library: 4
Distance from Depot to CSE Block: 5
Distance from Depot to Parking: 3
Distance from Hostel to Library: 3
Distance from Hostel to CSE Block: 4
Distance from Hostel to Parking: 5
Distance from Library to CSE Block: 2
Distance from Library to Parking: 4
Distance from CSE Block to Parking: 3
```

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```
Time from Hostel to Library: 2
Time from Hostel to CSE Block: 5
Time from Hostel to Parking: 4
Time from Library to CSE Block: 3
Time from Library to Parking: 3
Time from CSE Block to Parking: 4

========== OPTIMIZED SHUTTLE ROUTE ==========

Trip 1
Depot -> Hostel -> Library -> Depot

Trip 2
Depot -> CSE Block -> Parking -> Depot


Total Distance: 20.0 km
Total Travel Time: 23.0 minutes
Total Passenger Demand: 18
Bus Capacity: 10
Number of Trips: 2
Minimum Objective Cost: 43.0

All passenger demands are served.
```